AFCD Revision Notes

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Introduction

A course on social choice theory — a study of how individual prefernces can be combined, typically in the setting of voting.

This field is usually referred to as *Computational Social Choice*.

The bible for this field is the book *Handbook of Computational Social Choice* [**?**], see Chapters 1–4.

Also, Computational Social Choice (Felix Brandt, YouTube) was a basis for this course.

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Foundations

Definition 2.0.1. A binary relation R on a set U is:

- Complete: if $\forall x, y \in U$, xRy or yRx.
- Antisymmetric: if $\forall x, y \in U$, xRy and yRx implies $x = y$.

Notation 2.0.2. For $A \subset U$ and R a binary relation on U, we write $R|_A := R \cap (A \times A)$.

Definition 2.0.3 (Weak Ranking)**.** A weak ordering, or weak ranking on U is a binary relation on U that is reflexive, complete and transitive. (Can be thought of as a ranking on U with ties). Let $\mathcal{R}(U)$ denote the set of all weak rankings of U.

Notation 2.0.4. We use $\succsim$ to denote a weak ordering, and write:

- $a \succ b \iff a \succsim b \wedge \neg(b \succsim a)$
- $a \sim b \iff a \succsim b \wedge b \succsim a$

Definition 2.0.5 (Strict Ranking)**.** A linear order, or strict ranking on U is a binary relation on U that is reflexive, complete, transitive and anti-symettric. We let $\mathcal{L}(U)$ denote the set of all strict rankings on U.

Intuition 2.0.6. We should think of an element of $\mathcal{L}(U)$ as placing each alternative in U at a rung from best to worst with no two sharing the same rung. In comparison, we should think of an element of $\mathcal{R}(U)$ as placing element of U at rungs from best to worst, where we allow placing multiple alternatives at the same rung (they are tied).

Observation 2.0.7. $\mathcal{L}(U) \subseteq \mathcal{R}(U)$.

Definition 2.0.8. For binary relation R on U and $\emptyset \neq A \subseteq U$,

$$\text{Max}(R, A) := \{a \in A : \forall b \in A, bRa \implies aRb\}$$

Proposition 2.0.9. *The following hold,*

- *If R is complete, then* $\text{Max}(R, A) = \{a \in A : \forall b \in A, aRb\}$
- $\text{Max}(R, A)$ *can be empty*
- *If $R \in \mathcal{R}(U)$, then* $\text{Max}(R, A) \neq \emptyset$
- *If $R \in \mathcal{L}(U)$, then* $\text{Max}(R, A)$ *is a singleton*

Proof. The proofs follow trivially,

- We show $\forall b \in A, (bRa \implies aRb) \iff \forall b \in A, aRb$.

  The $\impliedby$ direction is immediated. For $\implies$ take $b \in A$, then bRa or aRb by completeness. If bRa, then aRb by the antecedent. If aRb, then we are done.

- Consider the rock, paper, scissor relation.

- SFC $\text{Max}(R, A) = \emptyset$.

  By the first item of this proposition $\text{Max}(R, A) = \{a \in A : \forall b \in A, aRb\}$. Then, since $\text{Max}(R, A) = \emptyset$, for every $a \in A$, there is $b \in A$ such that $\neg(aRb)$. R is reflexive so $\neg(aRb)$ implies $b \neq a$. Also by completeness, bRa. But this implies that U is infinite (we use transitivity), contradicting the finiteness of U.

- From the above item, $\text{Max}(R, A) \neq \emptyset$. Take $a, b \in \text{Max}(R, A)$, then aRb and bRa. By antisymmetry, $a = b$.

□

Remark 2.0.10. Note we used the finiteness of U, which we assume throughout.

Definition 2.0.11 (Prefernces over Alternatives)**.** Let U be a finite set, which we call the *alternatives*. A prefernce relation on U is linear order on U.

Typically, we consider multiple preference relations, we usually use the notation $\succsim_i \in \mathcal{L}(U)$ to denote the prefernces of voter i.

Remark 2.0.12. It is not without loss of generality to assume that preferences are linear orders, it would not be unreasonable to consider preference relations as weak orders.

In this course we assume preference relations are linear orders.

Definition 2.0.13 (Preference Profile)**.** Let $N = \{1, \ldots, n\}$ is a finite set of *voters* (we assume $n \geq 2$).

We call $R_N = (\succsim_1, \ldots, \succsim_n) \in \mathcal{L}(U)^N$ a *preference profile* over U [1].

Definition 2.0.14 (Feasible Sets)**.** We notate the *feasible sets* $\mathcal{F}(U) := \mathcal{P}(U) \setminus \{\emptyset\}$.

Definition 2.0.15 (Aggregation Functions)**.** A *Social Choice Function (SCF)* maps a prefernce profile and a feasible set A to a non-empty subset of A. Formally, $f : \mathcal{L}(U)^N \times \mathcal{F}(U) \to \mathcal{F}(U)$, is a social choice function if $f(R_N, A) \subseteq A$ for all $R_N \in \mathcal{L}^N$ and $A \in \mathcal{F}(U)$.

A *social welfare function (SWF)* maps a preference profile to a weak ordering on U. Formally, $g : \mathcal{L}(U)^N \to \mathcal{R}(U)$.

Example 2.0.16 (Trivial SCF)**.** The trivial SCF is defined as $f(R_N, A) := A$ for all $R_N \in \mathcal{L}^N$ and $A \in \mathcal{F}(U)$.

Example 2.0.17 (Borda's Rule)**.** Given $|U| = m$, for a voter with linear order $\succsim_i$ and an alternative $a \in U$, define the *Borda score* as

$$s_i(a) := |\, \{b \in U : a \succsim_i b\} \,| - 1$$

i.e., the number of alternatives ranked strictly below a by voter i. The top-ranked alternative gets $m - 1$ points, the bottom gets 0.

The *total Borda score* is $S(a) := \sum_{i \in N} s_i(a)$.

[1]We N as the superscript (instead of n) because we think of R_N as a function from $N \mapsto \mathcal{L}(U)$.

As a SWF: Define $a \succsim^B b \iff S(a) \geq S(b)$. This is a weak ordering on U (reflexivity, completeness, and transitivity are inherited from $\geq$ on $\mathbb{R}$).

As a SCF: Given feasible set A, return

$$f(R_N, A) := \text{argmax}_{a \in A}\, S(a)$$

the alternatives in A with the highest total Borda score. This is non-empty (a finite set has a maximum) and clearly a subset of A.

Note that the scores are computed over all of U, not just A. One could instead recompute Borda scores restricted to A; the two versions can disagree.

This is referred to as the *Broad Borda's rule*, and the version that first restricts to the feasible set and then computes the Borda scores is called the *Narrow Borda's rule*.

§2.1 Desired Properties and Initial Theorems

Definition 2.1.1 (Basic Fariness Conditions)**.** Often we hope that our aggregation functions satisfy certain properties, two most fundamental are the following (defined for SCFs):

- Anonymity: the SCF is invariant under permutations of the voters. Formally, for all pairs $R_N, R'_N \in \mathcal{L}(U)^N$ such that there is a permutation $\pi : N \to N$ with $\succsim_i = \succsim'_{\pi(i)}$, and for all $A \in \mathcal{F}(U)$, we have $f(R_N, A) = f(R'_N, A)$.
- Neutrality: the SCF is well-behaved under relabelling of the alternatives. Formally, for all pairs $R_N, R'_N \in \mathcal{L}(U)^N$ such that there is a bijection $\sigma : U \to U'$ with $x \succsim_i y \iff \sigma(x) \succsim'_i \sigma(y)$ for all $i \in N$ and $x, y \in U$, and for all $A \in \mathcal{F}(U)$, $\sigma(f(R_N, A)) = f(R'_N, \sigma(A))$.

Definition 2.1.2 (Pareto Optimality)**.** Another desirable property is Pareto optimality, which requires that if all voters prefer a to b, then b should not be chosen. More specifically, given $R_N \in \mathcal{L}(U)^N$ and $x, y \in U$, we say:

- x *Pareto-dominates* y if $\forall i \in N$, $x \succsim_i y$.
- y is *Pareto-dominated* if there is some $x \in U$ that Pareto-dominates y.
- x is *Pareto-optimal* if there is no $y \in U$ that Pareto-dominates x.

A SCF f satisifies *Pareto optimality* if for all $R_N \in \mathcal{L}(U)^N$ and $A \in \mathcal{F}(U)$, $f(R_N, A) \subseteq \{x \in A : x \text{ is Pareto-optimal}\}$.

One can define a *SCF* $f_{\text{Pareto}}(R_N, A) := \{x \in A : x \text{ is Pareto-optimal}\}$ that returns all Pareto-optimal alternatives.

NB1: Pareto-optimal alternatives always exist, to see as such, pick an alternative, if it is Pareto-optimal, we are done, otherwise there is an alternative that Pareto-dominates it, consider this alternative, if it is Pareto-optimal, we are done, otherwise there is an alternative that Pareto-dominates it. Given that U is finite, all that remains is to show that we cannot cycle here, but this is clear since the Pareto-relation is transitive (and irreflexive).

NB2: Pareto-dominated alternatives need not exist, consider two voters with opposite preferences over two alternatives a and b.

Definition 2.1.3 (Resoluteness)**.** A SCF f is *resolute* if for all $R_N \in \mathcal{L}(U)^N$ and $A \in \mathcal{F}(U)$, $f(R_N, A)$ is a singleton.

Proposition 2.1.4. *In general, an anonymous and neutral SCF is not resolute.*

Proof. Counter-example, take $U = \{a, b\}$ and $N = \{1, 2\}$. Then the profile:

$$R_N = \begin{pmatrix} a & b \\ b & a \end{pmatrix}$$

forces any anonymous and neutral SCF to return $\{a, b\}$ for the feasible set $A = \{a, b\}$.

In more detail, suppose for contradiction that f is anonymous, neutral, and resolute. Then $f(R_N, A) = \{a\}$ or $\{b\}$. WLOG assume $f(R_N, A) = \{a\}$. Consider now the profile R'_N obtained by swapping the two voters, i.e.,

$$R'_N = \begin{pmatrix} b & a \\ a & b \end{pmatrix}$$

By anonymity, $f(R'_N, A) = \{a\}$. But also by neutrality, $f(R'_N, A) = \{b\}$. Contradiction. □

Theorem 2.1.5 (Moulin, 1983)**.** *For n votes and m alternatives, there exists an anonymous, neutral, resolute, and Pareto-optimal SCF if and only if n is not divisible by any $k \in [2, \dots, m]$.*

In fact, for n not divisible by any $k \in [2, \dots, m]$, then the instant runoff *social choice function is anonymous, neutral, resolute, and Pareto-optimal.*

Definition 2.1.6 (Strategyproofness)**.** A SCF f is *strategyproof* if voters are never better off by misrepresenting their preferences.

Formally, we extend $\succsim_i$ to singleton sets (only), so that $\{x\} \succ_i \{y\} \iff x \succ_i y$. And we say that a SCF f is *manipulable* (by voter i), if there exists $R_N, R'_N \in \mathcal{L}(U)^N$, $A \in \mathcal{F}(U)$, such that $\succsim_j = \succsim'_j$ for all $j \neq i$, and $f(R'_N, A), f(R_N, A)$ are singletons and $f(R'_N, A) \succ_i f(R_N, A)$.

A SCF is *strategyproof* if it is not manipulable.

Definition 2.1.7 (Monotonicty)**.** A SCF f is *monotone* if the promotion of an alternative a in some voter's preference cannot cause a to be demoted from the set of winners.

Formally, f is monotone if for all $R_N, R'_N \in \mathcal{L}(U)^N$ and $A \in \mathcal{F}(U)$, such that the following hold:

- $\succsim_j = \succsim'_j$ for all $j \neq i$
- $x \succsim y \iff x \succsim' y$ for all $x, y \in U \setminus \{a\}$
- $a \succsim_i x \implies a \succsim'_i x$ for all $x \in U \setminus \{a\}$

then if $a \in f(R_N, A)$, $a \in f(R'_N, A)$.

Moreover, we say f is *positive responsive* if for all $R_N, R'_N \in \mathcal{L}(U)^N$ and $A \in \mathcal{F}(U)$, such that the same conditions hold, then if $a \in f(R_N, A)$, $\{a\} = f(R'_N, A)$.

Intuition 2.1.8. Clearly, positive responsiveness implies monotonicity. It is often easy to satisfy monotonicity, but much harder to satisfy positive responsiveness. For example, plurality is monotone but not positive responsive.

Proposition 2.1.9. *For a resolute SCF f on two alternatives, the following are equivalent.*

1. *f is strategyproof*
2. *f is monotone*
3. *f is positive responsive*

Proof.
(2) $\iff$ (3): This biconditional is trivial since resoluteness and monotonicity together imply positive responsiveness, and positive responsiveness always implies monotonicity.

(2) $\implies$ (1): As is typical in these settings, we proceed by contrapositive. Suppose f is manipulable by voter i. There exists profiles $R_N, R'_N \in \mathcal{L}(U)^N$ differing only in voter i's preferences, and $A \in \mathcal{F}(U)$ such that $f(R'_N, A) \succ_i f(R_N, A)$ — resoluteness gives that $f(R'_N, A)$ and $f(R_N, A)$ are singletons. Since $f(R'_N, A) \succ_i f(R_N, A)$ then $R'_N \neq R_N$. Without loss of generality, assume that the i-th preference relation of R_N and R'_N are

$$\gtrsim_i = \begin{pmatrix} a & b \end{pmatrix} \qquad\qquad \gtrsim'_i = \begin{pmatrix} b & a \end{pmatrix}$$

respectively. Since $f(R'_N, A) \succ_i f(R_N, A)$, $f(R'_N, A) = \{a\}$ and $f(R_N, A) = \{b\}$. It is clear that R_N and R'_N agree on all preference relations but that of voter i. It is vacuous that the i-th preference relation agrees the those relations not involving b. It is clear that R'_N strictly promotes b compared to R_N for voter i. But, $b \notin f(R'_N, A)$, so f is not monotone.

(1) $\implies$ (2): Again, by contrapositive, assume f is not monotone. Then there are profiles $R_N, R'_N \in \mathcal{L}(U)^N$ and $A \in \mathcal{F}(U)$ such that $f(R_N, A) = \{a\}$ and $f(R'_N, A) = \{b\}$, and R_N and R'_N differ only in voter i strictly promoting a in R'_N compared to R_N. As such, if voter i misrepresents their preferences as in R_N, then they get b, which they prefer to a. Thus, f is manipulable by voter i, and not strategyproof. □

Remark 2.1.10. As mentioned, positive responsiveness is rather difficult to satisfy, and for this reason we often consider a weakening of it that only holds for feasible sets of size 2; and denote this as *positive responsiveness*$_2$ *(PR$_2$)*.

Notation 2.1.11. For alternatives, a, b and preference profile R_N, we write $N_{ab} := \{i \in N : a \gtrsim_i b\}$, and $n_{ab} := |N_{ab}|$.

Intuition 2.1.12. Even with anonymity, neutrality, and only two alternatives, there are still several choices of SCFs. For one, we can define a threshold percentage of the vote won to have a single winner.

We can do weird things, like picking the minority vote, or defining a function based on if n_{ab}, n_{ba} is prime or not or is odd or not.

To rule out these more obtuse SCFs, we need something like monotonicity, or strategyproofness — intuitively they give that the SCF *moves* in the right direction.

Remark 2.1.13. Informal arguments suffice for showing neutrality and anonymity for exams - we need not go to first principles.

§2.2 May's Theorem(s)

Theorem 2.2.1 (May's Theorem)**.** *For two alternatives, the only SCF that is anonymous, neutral, and positive responsive is plurality.*

Proof. Consider that for two alternatives a and b, any anonymous SCF f must be a function of n_{ab} and n_{ba} and the choice of feasible set A only (strictly speaking, we do not need n_{ba}).

Let $f(m) := f(n_{ab}(R_N), \{a, b\})$. If $f(0) = \{a\}, \{a, b\}$, then (for $m > 1$) we get $f(m) = \{a\}$, violating neutrality. Thus, (and by symmetry) $f(0) = \{b\}$ and $f(m) = \{a\}$.

By positive responsiveness, if $a \in f(k)$ then $f(l) = \{a\}$ for each $l > k$. Let $k^* := \min\{k : a \in f(k)\}$. Suppose $k^* > \lceil n/2 \rceil$, then by considering the profile with a and b swapped we contradict neutrality. Likewise, for $k^* < \lceil n/2 \rceil$ a similar argument holds.

For $k^* = \lceil n/2 \rceil$, it is clear that neutrality gives $f(k^*) = \{a, b\}$.

Gap: Not particularly confident I can reproduce proof.

□

Theorem 2.2.2. *For two alternatives, of all the SCFs that are anonymous, neutral, and monotonic, plurality is the most* decisive *(give the fewest ties).*

Remark 2.2.3. It is so nice, and well-axiomatically motivated to use plurality when we have only two alternatives, and thus, a natural idea is to use plurality even when we have more than two alternatives.

§2.3 Further Properties and Impossibility results

Notation 2.3.1. $\gtrsim_M$ is the *pairwise majority relation*, defined by $a \gtrsim_M b \iff n_{ab} \geq n_{ba}$.

Observation 2.3.2 (Condorcet Paradox). $\gtrsim_M$ may contain cycles. The Condorcet Profile is:

$$R_N = \begin{pmatrix} a & b & c \\ b & c & a \\ c & a & b \end{pmatrix}$$

and contains cycles

$$a \gtrsim_M b, b \gtrsim_M c, c \gtrsim_M a$$

Definition 2.3.3 (Transitive Rationalisability)**.** A SCF f is *transitively rationalisable* if for each $R_N \in \mathcal{L}(U)^N$ there is a weak ordering R on U such that $f(R_N, A) = \text{Max}(R, A)$ for all $A \in \mathcal{F}(U)$.

Remark 2.3.4. The Condorcet Profile is not transitively rationalisable. Typically, in many rational decision making theories transitivity is thought to be a cornerstone. The Condorcet Paradox shows that despite individuals making transitive choices, the group choice can be intransitive.

Theorem 2.3.5. *The Condorcet Paradox can be alternatively be phrased as an impossibility result:*

> *There does not exists a SCF that is anonymous, neutral, positive responsive$_2$, and transitively rationalisable.*

Sketch. The first three properties essentially force pairwise choices according to majority rule. But then a Condorcet Profile can be constructed, which, given we make pairwise choices according to majority rule, is not transitively rationalisable. □

Definition 2.3.6. A Condorcet Winner is an alternative a such that $a \succ_M b$ for all $b \neq a$. Clearly, if a Condorcet winner exists, it does uniquely.

Definition 2.3.7 (Independence of Infeasible Alternatives (IIA))**.** A SCF f satisfies (IIA) if, for all A, R_N, R'_N: $R_M|_A = R'_M|_A$ implies $f(R_N, A) = f(R'_N, A)$.

Example 2.3.8. Consider the following pair of profiles,

R_N :

z_1	z_2	z_3
a	y	y
b	a	x
c	c	z
x	b	b
y	x	a
z	z	c

R'_N :

z_1	z_2	z_3
x	y	c
c	x	b
y	a	a
z	b	y
b	z	x
a	c	z

When restricted to $A := \{x, y, z\}$, they are identical. If f satisfies IIA, then $f(R_N, A) = f(R'_N, A)$.

Observation 2.3.9. A natural question in the set-up, is: Do we really need to know all the preferences of voters over all alternatives to determine the winner? If the SCF satisfies IIA then we do not.

Theorem 2.3.10 (Arrow's Impossibility Theorem)**.** *There is no SCF that satisfies IIA, Pareto-Optimality, non-dictatorship, and transitive rationalisability when there are at least three alternatives and at least two voters.*

An alternative formulation is *every SCF that satisfies IIA, Pareto-optimality, and transitive rationalisability is a dictatorship.*

Most commonly, Arrow's Impossibility Theorem is stated for SWFs. To be precise, we should define the notions of IIA, Pareto-optimality, transitive rationalisability, and dictatorship for SWFs.

Definition 2.3.11 (Properties for SWFs)**.** Let g be a SWF. For $R_N, R'_N \in \mathcal{L}(U)^N$, let $\succsim := g(R_N)$ and $\succsim' := g(R'_N)$.

- g satisfies *independence of irrelevant alternative (IIA)* if for all R_N, R'_N, x, y, $R_N|_{\{x,y\}} = R'_N|_{\{x,y\}}$ implies $\succsim|_{\{x,y\}} \iff \succsim|_{\{x,y\}}$.
- g satisfies *Pareto optimality* if for all R_N and x, y, $\forall i \in N, x \succ_i y$ implies $x \succ y$.
- g is dictatorial if if there is some $i \in N$ such that for all R_N, $g(R_N) = \succsim_i$.

Remark 2.3.12. Note that in a sense the notion of dictatorship for SWFs is stronger than for SCFs since the dictator can dictate not just the winners but also the entire ordering output.

Below, we drop the transitive rationalisability condition, since that is baked into the notion of a SWF.

Theorem 2.3.13 (Arrow's for SWFs)**.** *Every SWF that satisfies IIA and Pareto optimality is a dictatorship for three or more alternatives.*

Proofs of these formulations of Arrow's Impossibility Theorem are omitted, but we sketch the idea for two voters and three alternatives in the SWF case. The following table gives in each cell a preference profile, the row represents the preferences of voter 1 and the column represents the preferences of voter 2.

We fill out the table using just Pareto-Optimality for now, clearly along the diagonal any Pareto-Optimal SWF gives the same ordering as the voters, and elsewhere we can at least denote that one alternative must strictly beat another one.

	abc	acb	bac	bca	cab	cba
abc	abc	ab, ac	ac, bc	bc	ab	
acb	ab, ac	acb	ac		ab, cb	cb
bac	ac, bc	ac	bac	bc, ba		ba
bca	bc		bc, ba	bca	ca	ba, ca
cab	ab	ab, cb		ca	cab	ca, cb
cba		cb	ba	ba, ca	ca, cb	cba

Now, consider the cases in the table in more detail. Consider the cell corresponding to row (a, b, c) and column (b, c, a). We can do case analysis here. First suppose that g gives $a \succ b$ in this cell. Then, by IIA, any cell where the row heading prefers a to b and the column heading prefers b to a must have $a \succ b$.

This, with transitivity fills out more of the table. We repeatedly apply this reasoning to some other case. Eventually, we see that we get a dictatorship, and this holds no matter what case we choose.

Remark 2.3.14. What now? Arrow's and variations on this theorem are prevalent, and so if we want these easy and desirable properties it seems we must give up. We spend much of the course finding ways to escaper from this impossibility:

- restricted domains of preferences
- replace rationalisability with a weaker condition (variable electorate conditions)
- weaken rationalisability and/or IIA

3

Domain Restrictions

Remark 3.0.1. Implicit in Arrow's Impossibility Theorem is that all preference profiles are possible, in the literature this assumption is called *universal domain*. In some contexts, it makes sense to restrict the domain of possible preference profiles, to allow for better axiomatic results.

Example 3.0.2. Trivial Example: Only preference profiles are those in which all voters vote identically.

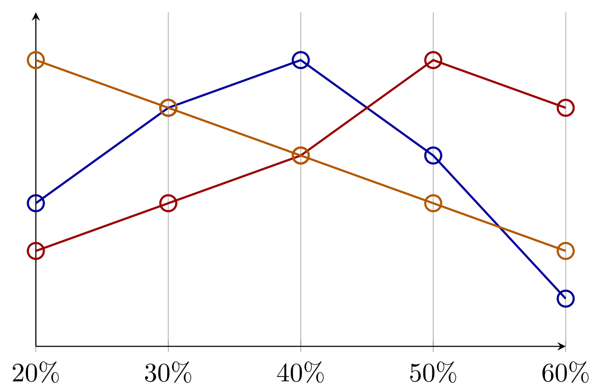
Single-peaked preferences are one of the most useful domain restrictions we can make. The following gives an example of what a single peaked preference is.

Example 3.0.3. Consider preferences over the maximal income tax rate. We have alternatives

$$U = \{20\%, 30\%, 40\%, 50\%, 60\%\}$$

and voters Alice, Bob, and Charlie with the following preferences:

○ Alice	○ Bob	○ Charlie
40%	50%	20%
30%	60%	30%
50%	40%	40%
20%	30%	50%
60%	20%	60%



Here, as is a reasonable assumption, the preferences of voters is single-peaked in some sense, according to the ordering of preference in the natural way. Of course, there is not usually a natural way to order preferences. We see more on single-peaked preferences later.

Definition 3.0.4. A restricted domain $\mathcal{D}(U)$ is a subset of $\mathcal{L}(U)^N$.

Example 3.0.5.

- The universal domain $\mathcal{D}_{\text{Univ}}(U) := \mathcal{L}(U)^N$.

- $\mathcal{D}_{\text{ab}}(U) = \{R_N \in \mathcal{L}(U)^N : (\forall i \in N, a \succ_i b) \vee (\forall i \in N, b \succ_i a)\}$ is the domain where all voters agree on the relative ranking of a and b.

Definition 3.0.6. A domain restriction $\mathcal{D}(U)$ is *Cartesian* if $\mathcal{D}(U) = \mathcal{L}'(U)^N$ for some $\mathcal{L}'(U) \subseteq \mathcal{L}(U)$.

In a Cartesian domain, the admissible preferences of a voter are independent of the preferences of other voters (which need not be true).

§3.1 Transitive Domains

Remark 3.1.1. $\mathcal{D}_{\text{ab}}(U)$ is not Cartesian, but $\mathcal{D}_{\text{Univ}}(U)$ is Cartesian.

Remark 3.1.2. As discussed, we use restricted domains to bypass impossibility results. In particular, we are interested in domain in which the majority relation $\succsim_M$ is transitive.

It is clear that in this restricted domain, $\succ_M$ contains no cycles, and so we get rid of the Condorcet Paradox.

In particular, if $\succsim_M$ is transitive, then $\text{Max}(\succsim_M, A)$ is a well-defined SCF.

Comment. We often like to think of (U, $\succsim_M$) as a graph.

Definition 3.1.3. Recall that an alternative $a \in U$ is a Condorcet winner w.r.t preference profile R_N if $a \succ_M b$ for all $b \neq a$.

We say that $a \in U$ is a weak Condorcet winner w.r.t to R_N if $a \succsim_M b$ for all $b \neq a$.

Proposition 3.1.4. *If $\succsim_M$ is transitive, then there exists a weak Condorcet winner.*

Proof. A transitive and complete relation on finite set has a maximum element. □

Remark 3.1.5. We thus, motivate the domain restricted to just those profiles with transitive majority relation, since we have a natural SCF that just outputs the weak Condorcet winners. This, couldn't be done in the general case since the Condorcet Paradox shows that we may have no weak Condorcet winners.

Definition 3.1.6. Let $\mathcal{D}_{\text{TRANS}}(U) := \{R_N \in \mathcal{L}(U)^N : \succsim_M \text{ is transitive}\}$.

As demonstrated if $\succsim_M$ is transitive then $\text{Max}(\succsim_M, A)$ is a well-defined SCF, and its elements are exactly the weak Condorcet winners, since they win or tie all pairwise majority comparisons.

Remark 3.1.7. $\mathcal{D}_{\text{TRANS}}(U)$ is not Cartesian.

Remark 3.1.8. In $\mathcal{D}_{\text{TRANS}}(U)$, the SCF $\text{Max}(\succsim_M, A)$ satisfies all the conditions of Arrow's Theorem.

It should be clear that it is not a dictatorship.

Consider that if everyone ranks one alternative over the other, then the majority must strictly (in fact trivially) rank that alternative over the other, and so clearly $\text{Max}(\succsim_M, A)$ satisfies Pareto optimality.

Consider that if A is a feasible set and R_N, R'_N are two profiles that are the same when restricted to A. Then consider that the pairwise majority in R_N and R'_N must be the same when restricted to A (clearly).

Finally, it is obvious that the majority relation is a transitive rationalisation of $\text{Max}(\succsim_M, A)$.

Lemma 3.1.9. $\text{Max}(\succsim_M, A)$ *is strategyproof in $\mathcal{D}_{TRANS}(U)$.*

Proof. Suppose for contradiction it is not. Then suppose that we have two profiles R_N, R'_N that differ only in voter i's preferences, and $A \in \mathcal{F}(U)$ such that $\text{Max}(\succsim'_M, A) \succ_i \text{Max}(\succsim_M, A)$ where $\text{Max}(\succsim'_M, A), \text{Max}(\succsim_M, A)$ are singletons. That is there is only one weak condorcet winner.

Let a be the weak Condorcet winner in R_N and a' be the weak Condorcet winner in R'_N. Thus, $a \succ_M a'$, $a' \succ'_M a$ (strictly because otherwise a' (resp. a) would also be weak condorcet winners, contradicting uniqueness), and $a' \succ_i a$. But then, consider that since $a \succ_M a'$, and $a' \succ_i a$, and R_N and R'_N differ only in voter i's preferences, the relative ranking in i's preferences between a and a' can only keep the same relative ranking or promote a compared to a'. Thus, it must be that $a \succ'_M a'$. Contradiction.

□

Corollary 3.1.10. *If $\mathcal{D}(U)$ is a subdomain of $\mathcal{D}_{TRANS}(U)$ (i.e. $\mathcal{D}(U) \subseteq \mathcal{D}_{TRANS}(U)$) then clearly the same proof above follows, and so we get that* $\text{Max}(\succsim_M, A)$ *is strategyproof in $\mathcal{D}(U)$.*

Remark 3.1.11. If there are an odd number of voters, then $\text{Max}(\succsim_M, A)$ is resolute in $\mathcal{D}_{\text{TRANS}}(U)$.

Proof. Clear. Cannot be that $n_{ab} = n_{ba}$. □

Theorem 3.1.12 (Cambell and Kelly, 2003)**.** *If n is odd,* $\text{Max}(\succsim_M, A)$ *is the only SCF that is non-dictatorial, Pareto-optimal, strategyproof, and resolute in $\mathcal{D}_{TRANS}(U)$.*

Proof. Omitted. □

§3.2 Single-Peaked Domains

We see that the domain $\mathcal{D}_{\text{TRANS}}(U)$ is a nice domain to work with, but perhaps it feels like a contrived example, we now turn to a natural example of a domain that is a subdomain of $\mathcal{D}_{\text{TRANS}}(U)$.

Definition 3.2.1. Let $\trianglelefteq \in \mathcal{L}(U)$ be a linear order of alternative. A preference relation $\succsim \in \mathcal{L}(U)$ is *single-peaked with respect to* $\trianglelefteq$ if for all $x, y, z \in U$: $((\mathrm{x} \triangleleft \mathrm{y} \triangleleft \mathrm{z})$ or $(\mathrm{z} \triangleleft \mathrm{y} \triangleleft \mathrm{x}))$ implies $(x \succ y \implies y \succ z)$.

This says, that if the linear order ranks y between x and z, then if the preference relation ranks x over y, then it must rank y over z. This gives a kind of monotonicity condition everywhere except for when the linear order ranks y at the top of x and z, in which case there is no restriction, and y may be the peak.

Alternatively, consider what this definition forbids. That is, it disallows the middle alternative from being ranked lowest (think this over).

Definition 3.2.2. For a linear order $\trianglelefteq \in \mathcal{L}(U)$, and a preference profile $R_N \in \mathcal{L}(U)^N$, we say that R_N is *single-peaked with respect to* $\trianglelefteq$ if each preference relation in R_N is single-peaked with respect to $\trianglelefteq$.

Definition 3.2.3. We define the *domain of single-peaked preferences* with respect to $\trianglelefteq$ as

$$\mathcal{D}_{\text{SP}}^{\trianglelefteq}(U) := \left\{ R_N \in \mathcal{L}(U)^N : R_N \text{ is single-peaked with respect to } \trianglelefteq \right\}$$

Equivalently, we can define it as

$$\mathcal{D}_{\text{SP}}^{\trianglelefteq}(U) := \{ \succsim \in \mathcal{L}(U) : \succsim \text{ is single-peaked with respect to } \trianglelefteq \}^N$$

Clearly, under this definition we see that $\mathcal{D}_{\text{SP}}^{\trianglelefteq}(U)$ is Cartesian.

Remark 3.2.4. Single-peaked arises naturally in a few situations,

- The desired distance of a train station from your house
- Left-Right political spectrum

- Age for people to be allowed to vote

The big motivation for considering single-peaked preferences, is the following theorem, which follows from the previous section.

Theorem 3.2.5 (Black, 1948). *$\succsim_M$ is transitive in $\mathcal{D}^{\trianglelefteq}_{SP}(U)$.*

- *In particular, if n is odd, every single-peaked profile admits a (strict) Condorcet winner.*
- *For even n, weak Condorcet winners are guaranteed*

We defer the proof to ???, from which it follows trivially.

Remark 3.2.6. The above also holds for so-called, *single-caved* preferences $\mathcal{D}^{\trianglelefteq}_{\mathrm{SC}}(U)$. A preference relation $\succsim$ is single-caved with respect to $\trianglelefteq$ if for all $x, y, z \in U$: $((\mathrm{x} \triangleleft \mathrm{y} \triangleleft \mathrm{z})$ or $(\mathrm{z} \triangleleft \mathrm{y} \triangleleft \mathrm{x}))$ implies $(y \succ x \implies z \succ y)$.

Theorem 3.2.7. *We give some intuition as to why $\succsim_M$ is transitive in $\mathcal{D}^{\trianglelefteq}_{SP}(U)$.*

Assume that n is odd, let t_i denote the alternative ranked highest by voter i (the top-choice *of voter i). We can imagine placing pebbles along the linear order $\trianglelefteq$ for each voter according to their top choice. Then the alternative that the median pebble (according to $\trianglelefteq$) is placed on is the Condorcet winner.*

Proof. Gap: TODO: Easy proof.

□

Remark 3.2.8. This proves something that is just a slightly weaker Black's theorem.

Remark 3.2.9 (Median Voting). By the above, that just the voters' top choices are sufficient to compute the Condorcet winner (when one exists).

> Importantly, we note that the plurality winner may differ from the Condorcet winner, even in $\mathcal{D}^{\trianglelefteq}_{\mathrm{SP}}(U)$.

In the domain $\mathcal{D}^{\trianglelefteq}_{\mathrm{SP}}(U)$ this — which is just $\mathrm{Max}(\succsim_M, A)$ — is known as *median voting*.

Median voting satisfies strategyproofness in the domain $\mathcal{D}^{\trianglelefteq}_{\mathrm{SP}}(U)$. This follows from Black's theorem and Lemma 3.1.9.

In $\mathcal{D}^{\trianglelefteq}_{\mathrm{SP}}(U)$, it is computationally trivial to find the Condorcet winner. So far, we have not thought much about the computational social choice theory from a complexity perspective, but this is a theme of the course that we will now be seeing more of.

Remark 3.2.10. Can we efficiently check whether a preference profile is single-peaked with respect? If we are given some order $\trianglelefteq$, we can trivially check whether a preference profile is single-peaked with respect to $\trianglelefteq$ by verifying the single-peaked condition for each voter's preferences We check the single-peaked condition for each preference relation in a profile by going through each voter's contiguous 3-tuples of preferences (think!) i.e. in time linear in the number of alternatives. Thus, given $\trianglelefteq$, we can check whether a profile is single-peaked with respect to $\trianglelefteq$ in time $O(n|U|)$.

The more interesting question is whether we can efficiently check single-peakedness without being given $\trianglelefteq$; after all, we have all these nice properties if a profile is single-peaked with respect to some order, so it is nice to be able to easily check whether a profile is single-peaked. Of course, checking all linear orders is not efficient.

Observation 3.2.11. A least-ranked alternative in a preference relation must be placed at the leftmost or rightmost end of the linear order.

Gap: Reason why this is true?

Thus, there can be at most two least-ranked alternatives. This motivates the idea that we try to arrange the alternatives from the outside to the inside.

Idea 3.2.12. From this, we attempt to construct a linear order by placing the least-ranked alternatives at the ends, and iterating this. Suppose that after one iteration l and r are the least-ranked alternatives in the profile and so they get placed at the innermost left and right positions of the partially constructed linear order. Then, we remove l and r from the profile and repeat this process. Let $A \subseteq U$ be the set of alternatives that have not yet been arranged.

Suppose that x is the bottom ranked alternative in the profile restricted to A (so needs to be placed next). If there is a voter that ranks $l \succ_i x \succ_i r$, and that voter does not rank x as their top choice when we restrict to A, then we have to place x on the right (next to r).

Why?

Suppose for contradiction that x is placed on the left, just inside l. Then in $\trianglelefteq$ the arrangement is:

$$\cdots \quad l \quad x \quad \underbrace{A \setminus \{x\}}_{\ni t_i} \quad r \quad \cdots$$

where t_i denotes the peak of voter i restricted to A, which is not x by assumption. Since $t_i \in A \setminus \{x\}$, it lies strictly to the right of x in $\trianglelefteq$. Moving leftward from t_i, we reach x before l, so x is closer to t_i than l is. Single-peakedness then requires $x \succ_i l$. But voter i has $l \succ_i x$, a contradiction. Thus x must be placed on the right, next to r.

Gap: Check this makes sense.

Algorithm 1 Single-Peaked Recognition

Set dummy leftmost alternative to $z_l \notin U$, rightmost to $z_r \notin U$
$l \leftarrow z_l$, $r \leftarrow z_r$
$A \leftarrow U$
while $|A| \geq 2$ **do**
    $B \leftarrow \{x \in A : x \text{ is bottom-ranked in } R_N|_A \text{ for some voter}\}$
    $R \leftarrow \{x \in B : \exists i \in N \text{ such that } l \succ_i x \succ_i r \text{ and } t_i(A) \neq x\}$
    $L \leftarrow \{x \in B : \exists i \in N \text{ such that } r \succ_i x \succ_i l \text{ and } t_i(A) \neq x\}$
    if $|B| \leq 2 \wedge |L| \leq 1 \wedge |R| \leq 1 \wedge L \cap R = \emptyset$ **then**
        Place the alternative in L (if any) next to l; update l
        Place the alternative in R (if any) next to r; update r
        Place any $x \in B \setminus (L \cup R)$ in an arbitrary empty slot next to l or r
    else
        return R_N is not single-peaked
    end if
    $A \leftarrow A \setminus B$
end while
if $|A| = 1$ **then**
    Place the remaining $x \in A$ in the last slot
end if
return R_N is single-peaked with the constructed order $\trianglelefteq$

Gap: TODO: Write and understand this in more detail, also trace this algorithm for an example.

Theorem 3.2.13. *The above algorithm is correct and runs in time $O(n|U|)$ — so in time that is linear in in the size of a specification of a preference profile.*

Proof. See [**?**][Section 4.1.]. □

Remark 3.2.14. Essentially the same algorithm can be used to check whether a profile is single-caved, by running it on a preference profile that is flipped upside down.

Gap: TODO: Important, re-watch last 10mins of 03/02/2026 — make notes on this section.

4

Borda, Condorcet, and Kemeny

Idea 4.0.1. The *theme* of this chapter comes from the conflict between the social choice ideas of Borda and Condorcet.

Definition 4.0.2. In this chapter, we consider a variable set of voters.

Formally speaking, we consider the set of preference profiles over $\bigcup_{N \subseteq \mathbb{N}; |N| < \infty} \mathcal{L}(U)^N$. This allows us to consider profiles with different numbers of voters, and consider merging profiles $R_N \in \mathcal{L}(U)^N$ and $R_{N'} \in \mathcal{L}(U)^{N'}$ into a profile $R_N \cup R_{N'} \in \mathcal{L}(U)^{N \cup N'}$ (assuming $N \cap N' = \emptyset$).

Definition 4.0.3 (Scoring Rule)**.** Informally speaking, for a fixed feasible set $A \subseteq U$, a *score vector* $s := (s_1, s_2, \ldots, s_{|A|})$ is a real vector, such that if voter ranks an alternative at the i-th position, then it gets s_i points. A *scoring rule* is a SCF that choses those alternatives with the highest total score.

More formally, we should define the notion of a score vector for each possible size of feasible set. Let $s^{(1)}, s^{(2)}, \ldots, s^{(|U|)}$ be a collection of score vectors where $s^{(k)}$ has size k. The corresponding scoring rule is defined as $\operatorname{argmax}_{x \in A} s(x, A)$, where,

$$s(x, A) := \sum_{i \in N} s^{(|A|)}_{|y \in A : y \succsim_i x|}$$

Remark 4.0.4. The above definition is a bit weird/annoying, it allows for perversities like Borda for even sized feasible sets and Plurality for odd sized feasible sets. Usually, we think about the feasible set as fixed and often omit it from the notation.

Example 4.0.5.

- Borda's rule, $s^{(|A|)} := (|A| - 1, |A| - 2, \ldots, 0)$.

  Gap: Think if this is broad or narrow Borda?

- Plurality, $s^{(|A|)} := (1, 0, \ldots, 0)$
- *Anti-Plurality*, $s^{(|A|)} := (1, 1, \ldots, 1, 0)$

Proposition 4.0.6. *Scoring rules are invariant under* positive affine transformations [1] *of their score vectors.*

Proof. Easy boring exercise. ∎

[1] $s_i^{(k)} \mapsto \alpha_i s_i^{(k)} + \beta_i$ where $\alpha_i > 0$.

Proposition 4.0.7. *Scoring rules can be efficiently computed* [2].

Proof. We compute them in the obvious way. ☐

Proposition 4.0.8. *A scoring rule is monotonic if and only if for all k, $s_1^{(k)} \geq s_2^{(k)} \geq \ldots \geq s_k^{(k)}$.*

A monotonic scoring rule is non-trivial *if $s_1^{(k)} > s_k^{(k)}$.*

Proof. Exercise. ☐

Definition 4.0.9 (Reinforcement)**.** A SCF f satisfies *reinforcement* if for all A and disjoint sets of voters N and N', and all $R_N \in \mathcal{L}(U)^N$ and $R_{N'} \in \mathcal{L}(U)^{N'}$, such that $f(R_N) \cap f(R_{N'}) \neq \emptyset$,

$$f(R_N \cup R_{N'}, A) = f(R_N, A) \cap f(R_{N'}, A)$$

Informally, whenever there is a common alternative selected by two disjoint sets of voters (under the SCF), then the common alternatives should be the only alternatives selected when we merge the two sets of voters together.

Example 4.0.10. Let's think about which rules satisfy this. It is clear that Plurality satisfies this.

Gap: Think why? Also, does Borda satsify it?

Theorem 4.0.11 (Smith, 1973; Young, 1975)**.** *A neutral and anonymous SCF satisfies reinforcement if and only if it is a* composed *scoring rule.*

Definition 4.0.12 (Composed scoring rule)**.** An SCF is a *composed* scoring rule if there is (finite) sequence of scoring rules where each scoring rules (if there is a next scoring rule) is used to break ties, (if there are any), of the previous rule.

Any scoring rule is a composed scoring rule, by taking the length of the sequence to be one.

Intuition 4.0.13. Roughly speaking, reinforcement characterises scoring rules (composed) — sometimes reinforcement can be used to reason about all scoring rules.

Definition 4.0.14 (Cancellation)**.** An SCF satisfies *cancellation* if for all A and R_N,

$$\left(\forall x, y \in A : n_{xy} = n_{yx}\right) \implies f(R_N, A) = A$$

NB: Cancellation is only meaningful when n is even.

Remark 4.0.15. Borda is a scoring rule which satisfies cancellation. To see as such, consider we can alternatively formulate the total Borda score $s_{\text{NarrowBorda}}(x, A)$ of alternative x as, the sum of the number of voters that rank $y \neq x$ below x for each alternative y, i.e.,

$$s_{\text{NarrowBorda}}(x, A) = \sum_{y \in A \setminus \{x\}} n_{xy}$$

This is just changing the order of summation from summing first over alternatives and then over voters to the reverse order. As such, we see that if for all $x, y \in A$, $n_{xy} = n_{yx}$, then clearly $s_{\text{NarrowBorda}}(x, A) = s_{\text{NarrowBorda}}(y, A)$ for all $x, y \in A$, and so $f(R_N, A) = A$.

Remark 4.0.16. Plurality does not satisfy cancellation. Consider the profile on three alternatives with two voters where one voter has preferences $a \succ_1 b \succ_1 c$ and the other has the reverse $c \succ_2 b \succ_2 a$. Then, for the total feasible set $A = \{a, b, c\}$, $n_{ab} = n_{ba} = 1$, $n_{ac} = n_{ca} = 1$, and $n_{bc} = n_{cb} = 1$, so the cancellation condition is satisfied, but $f(R_N, A) = \{a, c\} \neq A$.

Theorem 4.0.17 (Young, 1974)**.** *Borda's (narrow) rule is the only SCF satisfying neutrality, Pareto-optimality, reinforcement, and cancellation.*

[2] Meaning it is in **P**.

§4.1 Family of Condorcet Extensions

Definition 4.1.1. An SCF f is a *Condorcet extension* if it picks a the (strict) Condorcet winner whenever there is one, and otherwise does something (anything) else.

Gap: Can you think of a SCF which is a Condorcet extension?

4.1.1 Fishburn's Classification

Idea 4.1.2. Fishburn's Classification is used to categorise Condorcet Extensions based on the amount of *information* used to compute them.

Definition 4.1.3 (C1 Graph)**.** The *C1 Graph* or *Majority Graph* with respect to a preference profile R_N is the directed graph with vertex set U and and edge from a to b if $a \succ_M b$. That is, the edges are given by the majority relation, where there is an edge from a to b iff a strictly wins the pairwise majority comparison with b.

Definition 4.1.4 (C2 Graph)**.** The *C2 Graph* or *Weighted Majority Graph* is the directed graph with vertex set U and edges from a to b with weight n_{ab}.

Remark 4.1.5. It is clear that we can extract the C1 Graph from the C2 Graph by just looking at the edge weights and seeing whether an edges has weight greater than the reverse edge.

Definition 4.1.6 (Margin Graph)**.** The *Margin Graph* is the directed graph with vertex set U and an edge from a to b with weight $n_{ab} - n_{ba}$ whenever $n_{ab} > n_{ba}$.

Remark 4.1.7. It is clear that the margin graph has precisely the same edges as the C1 graph, but with weights given by the margin of victory.

Definition 4.1.8 (Fishburn's Classification)**.** Fishburn classifies Condorcet extensions (and more generally SCFs) into three classes C1, C2, and C3.

- **C1:** f depends only on the majority relation, that is it is a function of the C1 Graph.
- **C2:** f depends only on the weighted majority relation and is not in C1, that is it is a function of the C2 Graph (but not just the C1 Graph).
- **C3:** f is not a function of the C2 Graph (and hence it is not C1 nor C2).

Example 4.1.9 (Copeland's Rule)**.** We define Copeland's Rule as,

$$f_{\text{Copeland}}(R_N, A) := \text{argmax}_{x \in A} \left|\{y \in A : x \succ_M y\}\right|$$

That is, the Copeland rule gives those alternatives that win the most pairwise majority comparisons.

The Copeland rule is a C1 rule, since it depends only on the majority relation. We can see that the Copeland rule can be thought of as returning all those alternatives that have the maximal outgoing edges in the C1 graph.

We also note, that the Copeland Rule is a Condorcet extension, since if there a Condorcet winner, i.e. an alternative that wins all pairwise majority comparisons, then it is the unique alternative with the most outgoing edges in the C1 graph, and so it is the unique winner under the Copeland Rule.

Example 4.1.10 (Maximin Rule)**.** We define the Maximin rule as,

$$f_{\text{Maximin}}(R_N, A) := \text{argmax}_{x\in A} \min_{y\in A\setminus\{x\}} n_{xy}$$

That is, the Maximin rule gives those alternatives whose weakest pairwise majority victory is as strong as possible.

The Maximin rule is a C2 rule, since it depends on the weighted majority relation, but not just the majority relation. We can see that the Maximin rule can be thought of as returning all those alternatives that have the maximal weight on their weakest outgoing edge in the C2 graph.

Let us pause to think how we would formally show that it is a C2 rule. We would exhibit profiles that have identical C1 Graphs, but for which the maximin rule differs.

We also note, that the Maximin Rule is a Condorcet extension, since it must win in the majority relation, so if there is a Condorcet winner a, it must be that $n_{ay} > n_{ya}$ for all $y \neq a$, so $n_{ay} > n/2$. For any other alternative $x \neq a$, then since $n_{ax} > n/2$ and there are n voters, it must be that $n_{xa} < n/2$.

Example 4.1.11 (Borda's (narrow) Rule)**.** We note that, Borda's (narrow) rule is given by,

$$f_{\text{Borda}}(R_N, A) := \text{argmax}_{x\in A} \sum_{y\in A\setminus\{x\}} n_{xy}$$

Thus clearly it is a C2 rule, since it depends on the weighted majority relation. On the C2 graph, we compute the Borda winners as those alternatives that have the maximal total outgoing edge weight.

It should be noted that Borda's rule is **not** a Condorcet extension.

Gap: Think why?

Example 4.1.12. Plurality is a C3 rule, since it is not a function of the C2 graph.

Gap: Come up with two profiles with identical C2 Graphs but differing Plurality winners

Gap: Aside from Borda's rule what other (if any) scoring rules are in C1, C2, and C3

.

Example 4.1.13 (Young's Rule)**.** Returns all alternatives that can be made a (strict) Condorcet Winner when we remove a minimal number of voters. That is, over each alternative, what is the fewest number of voters we need to remove to make that alternative a Condorcet winner, and then we return those alternatives for which this number is minimal.

Young's rule is obviously a Condorcet extension. Young's rule is intractible. Young's rule is a C3 rule, for some intuition for why it is not C2, to know what voters to delete, we need to have access to the rankings in the profile (try to prove it formally).

Remark 4.1.14. When there are only two alternatives, majority rule (which is just plurality) is the the only non-trivial, monotonic scoring rule, and the only Condorcet extension.

Proposition 4.1.15 (Condorcet, 1785)**.** *Borda's rule is not a Condorcet extension when there are three or more alternatives.*

Proof. First for the case of three alternatives consider the profile given by four voters having preferences $a \succ b \succ c$, and three voters having $b \succ c \succ a$. Then a gets a total Borda score of $4 \cdot 2 + 3 \cdot 0 = 8$, and b gets a total Borda score of $4 \cdot 1 + 3 \cdot 2 = 10$. So Borda's rule does not select a. Despite this, a is the Condorcet winner, since a beats b and c by 4 votes to 3.

Now in general suppose there are $m \geq 3$ alternatives and consider the profile given by n_1 voters having preference relations $P_1 : a \succ b \succ \cdots$, and n_2 voters having preference relations $P_2 : b \succ \cdots \succ a$. For a to be a Condorcet winner, we need $n_1 > n_2$. Consider that a gets a total Borda score of $n_1 \cdot (m-1) + n_2 \cdot 0 = n_1(m-1)$, and b gets a total Borda score of $n_1 \cdot (m-2) + n_2 \cdot (m-1)$. We can thus ensure that b has a higher Borda score than a and that a is the Condorcet winner by choosing n_1 and n_2 such that $n_2 < n_1 < (m-1)n_2$, for $m \geq 3$ this can be achieved easily.

Hence, we can construct a profile where a is the Condorcet winner but b has a higher Borda score, and so Borda's rule does not select the Condorcet winner for each $m \geq 3$ (observe the inequality fails when $m = 2$). The above example for $m = 3$ does exactly this. □

Theorem 4.1.16. *No scoring rule is a Condorcet extension when there are three or more alternatives.*

Proof. Exercise. (Hint: Condition first on monotonic and non-monotonic scoring rules, maybe use Proposition 4.0.6 and then construct profiles.) □

Remark 4.1.17. We could think of Borda scoring as giving points per rank, and describe Borda's rule as choosing the alternatives with the highest average rank.

Theorem 4.1.18 (Smith, 1973)**.** *A Condorcet winner is never the alternative with the lowest Borda score. Borda's rule is the only scoring rule for which this holds.*

We can think of this as saying that Borda is the scoring rule which is most compatible with the Condorcet criterion.

Equivalently, a Condorcet loser [3] *is never the alternative with the highest Borda score (and again the only scoring rule for which this works is Borda's).*

Proof. Exercise. □

Theorem 4.1.19 (Gehrlein et al., 1978)**.** *When all preferences profiles are equally likely, over all scoring rules, Borda's rule maximizes the probability of selecting the Condorcet winner (when one exists).*

Idea 4.1.20 (Unifying Borda and Condorcet)**.** Despite there being conflict between Borda and Condorcet there have been some attempts to unify the rules.

- Black's Rule: Return the Condorcet winner when one exist, otherwise the Borda winner(s) (by definition a Codorcet extension). - it is slightly ad hoc.
- Baldwin's Rule: Run-off method based on (narrow) Borda's rule. In each round, all the minimal (narrow) Borda score alternatives are deleted, until no more deletions are possible — output this set now. Baldwin's rule is a Condorcet extension (Exercise), but it is not monotonic [4].

Theorem 4.1.21 (Young and Levenglick, 1978)**.** *For three or more alternative, no Condorcet extension satisfies reinforcement*

Proof. Assume for contradiction that f is an SCF which is a Condorcet extension satisfying reinforcement. Consider the profile R_N given below,

$a \succ b \succ c$	$b \succ c \succ a$	$c \succ a \succ b$
2 voters	2 voters	2 voters

[3]An alternative x is a Condorcet loser if it strictly loses all non-reflexive pairwise majority comparisons.
[4]Every scoring run-off rule violates monotonicity (Smith, 1973)

Since $f(R_N, \{a, b, c\})$ must select some alternative and the profile is symmetric in a, b, and c, there is no Condorcet winner, so without loss of generality assume $b \in f(R_N, \{a, b, c\})$.

Consider the profile $R_{N'}$ below,

$b \succ a \succ c$	$a \succ b \succ c$
2 voters	1 voter

Then, in $R_{N'}$, b is the Condorcet winner, so $f(R_{N'}, \{a, b, c\}) = \{b\}$. By reinforcement, $f(R_N \cup R_{N'}, \{a, b, c\}) = f(R_N, \{a, b, c\}) \cap f(R_{N'}, \{a, b, c\}) = \{b\}$.

Now, consider the combined profile, here a is the Condorcet winner, so $f(R_N \cup R_{N'}, \{a, b, c\}) = \{a\}$. Contradiction.

It is convenient in this proof to draw the margin graphs for R_N and $R_{N'}$, then we can easily combine them to get the margin graph for $R_N \cup R_{N'}$. □

Idea 4.1.22. To prove this for a general number of alternatives, we can *cheat* by adding extra preferences at the bottom of the profiles R_N and $R_{N'}$, and taking the same feasible set. Exercise: can we show that the theorem holds for the total feasible set on any number of alternatives greater than three.

Usually, when something goes wrong for some number of alternatives, it does for larger numbers of alternatives too — so it is interesting when it first goes wrong, which is usually for three alternatives (majority is axiomatically nice for two alternatives).

Idea 4.1.23 (Social Preferences)**.** SPF : $\mathcal{L}(U)^N \to \mathcal{F}(\mathcal{L}(U))$, where $\mathcal{F}(\mathcal{L}(U))$ is the set of non-empty finite subsets of $\mathcal{L}(U)$.

In this context, Young and Levenglick's theorem (Theorem 4.1.21) doesn't hold.

§4.2 Kemeny's Rule

Definition 4.2.1 (Social Preference Functions)**.** A *social preference function* (SPF) is a function h that maps a preference profile to a non-empty set of linear orders over U. That is,

$$h : \mathcal{L}(U)^N \to \mathcal{F}(\mathcal{L}(U))$$

We can think of a social preference function as a generalisation of a social welfare function, where we can output a set of orders instead of just one, and where the orders must be linear orders instead of just weak orders.

Definition 4.2.2 (Kemeny's Rule)**.** Kemeny's rule is a SPF defined as,

$$h_{\text{Kemeny}}(R_N) := \text{argmax}_{\succsim \in \mathcal{L}(U)} \sum_{i \in N} |\{\langle x, y\rangle : x \succ y, x \succ_i y\}|$$

As with SCFs, we approach them from an optimization perspective. Here, instead of optimizing over alternatives, we optimize over linear orders.

Informally, for each linear order, we give it points based on how much it agrees with a voters preference, and sum over all voters, and then we return (all) those linear orders that have the highest total score.

NB: We can think about Kemeny's rule as that which yields all rankings that maximize agreement with the profile. More intuitively, we can think about it minimizing the pairwise disagreement as measured by the bubble-sort distance.

Example 4.2.3. For some linear order say $a \succ b \succ c$, we can compute the Kemeny-Score as,

$$\text{Kemeny-Score}(\begin{pmatrix} a \\ b \\ c \end{pmatrix}) = n_{ab} + n_{ac} + n_{bc}$$

It is clear that there is an exponential number of linear orders, thus an exponential number of Kemeny-Scores we must compute, so it is not the easiest to compute Kemeny's rule by brute-force.

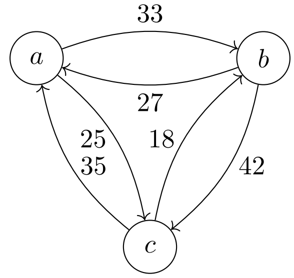
Also, note that in some sense (despite it being a SPF) Kemeny's rule only relies on the C2 graph, since we can compute the Kemeny-Score of a linear order just by looking at the edge weights in the C2 graph.

We can think of Kemeny's rule as picking the maximal weight acyclic selection of direction on all pairs of the edges [5]. What if we didn't care for acyclicity? We would then always pick the direction of the edge with the higher weight, and in some sense this gives us an *ideal* score, upper bounding the Kemeny-Score — that is, if cyclic rankings were *allowed*, $\succsim_M$ would maximize the Kemeny-Score.

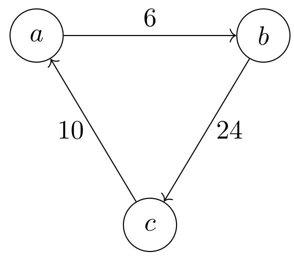
We can think about starting with the majority cycle (described above) and then massaging this into a linear order by changing the direction of some edges, any change invokes a penalty in the Kemeny-Score of $n_{xy} - n_{yx}$, precisely the the edge weight in the margin graph.

Example 4.2.4. Consider the following profile, given with its C2 graph below, and its margin graph below that.

23	17	10	8	2
a	b	c	c	b
b	c	a	b	a
c	a	b	a	c



[5]An acyclic tournament is a linear order.



We look at the C2 graph, and notice that the *ideal* majority score (when we disregard the acyclicity condition) is given by $33 + 42 + 35 = 110$.

We can consider the linear order $a \succ b \succ c$, and observe (via the margin graph) that this incurs a penalty of 10 for the edge $a \succ c$. It is not too hard to see, that this means that we get the Kemeny-Score of $110 - 10 = 100$ for this linear order. Perhaps, it is not too much of a stretch to notice that this is not the highest Kemeny-Score we can get, since we can instead incur the 6 penalty for the edge $a \succ b$ by considering the linear order $b \succ c \succ a$.

Remark 4.2.5. We can then reduce the problem of computing the Kemeny winners, to a problem in the margin graph. Namely, we find sets of edges that are minimal in total weight, and such that the graph produced by reversing these edges (and keeping the rest of the margin graph as is) is acyclic.

Idea 4.2.6. This idea of picking a subset of edges, reversing the direction of that subset, and then checking acyclicity of the resulting graph, is a little bit of a pain to think about. Instead, we would like to think about deleting these edges instead, and the following lemma helps with this.

Lemma 4.2.7. *Let $G = (V, E)$ be a directed graph, and $E' \subseteq E$ be a subset of edges. Then, G can be made acyclic by inverting a subset $E'' \subseteq E'$ iff $(V, E \setminus E')$ is acyclic.*

Proof.
$\Longrightarrow$: Easy, remove all of E'.

$\Longleftarrow$: Greedily insert edges in E' back in, one after another. At each step orient the edges so that no cycle forms. If at any stage, both orientations yield a cycle, then the previous graph was cyclic. Contradiction. □

Comment. This formulation is not as clean as we may hope, but is necessary, for example, consider the three cycle graph, deleting the whole cycle makes is acyclic, but reversing it does not.

Remark 4.2.8. This lemma, allows us to consider the problem now to be, find a subset of edges E' with minimal total weight, such that $(V, E \setminus E')$ is acyclic.

We might hope this to now be rather easy, "For each cycle, remove the minimum weight edge.", but cycles may overlap, and so this is slightly more tricky.

Comment. Recall, we showed that reinforcement is satisfied by all scoring rules, and no Condorcet extensions — redefining these properties for SPFs, we can show that there is a Condorcet extension that satisfies reinforcement, namely, Kemeny's rule (cheating, of course by redefining the properties in this context).

Definition 4.2.9 (Young's Characterization)**.**
An SPF h satisfies *reinforcement* if for all disjoint sets of voters N and N', and all profiles $R_N \in \mathcal{L}(U)^N$ and $R_{N'} \in \mathcal{L}(U)^{N'}$, where $h(R_N) \cap h(R_{N'}) \neq \emptyset$, then $h(R_N \cup R_{N'}) = h(R_N) \cap h(R_{N'})$ [6].

Two alternatives x and y are *adjacent* in a preference relation $\gtrsim$ if there is no alternative z such that $x \succ z \succ y$ or $y \succ z \succ x$. We say an SPF h satisfies *Condorcet-consistency* if for all profiles R_N, $\gtrsim \in h(R_N)$, and $x \succ y$

[6]Again, formally, we are considering this notion of an expanded domain.

where x and y are adjacent in $\gtrsim$, we have $x \succ_M y$. Moreover, $x \sim_M y$ implies $(\gtrsim \setminus \{\langle x, y\rangle\}) \cup \{\langle y, x\rangle\} \in h(R_N)$ [7].

Remark 4.2.10. Consider that Condorcet-consistency, implies the natural notion that we might think of as a Condorcet extension in the context of SPFs. Namely, if there is a Condorcet winner and an SPF is Condorcet-consistent, then the Condorcet winner is ranked above all other alternatives in every linear order given by the SPF.

Gap: Why?

Theorem 4.2.11 (Young and Levenglick, 1978)**.** *Kemeny's Rule is the only neutral SPF satisfying reinforcement and Condorcet-consistency.*

Remark 4.2.12. Without going into the proof of the *only* part of the above theorem, it is a good exercise to think why Kemeny's rule is neutral and satisfies reinforcement and Condorcet-consistency.

Comment. Often, we want to reason in the course about profiles by way of their C1, C2, or margin graphs. It is often easier to go straight from a graph, and we may not first want to construct a profile which yields that graph.

Theorem 4.2.13 (McGarvey's Theorem)**.** *For tournament $G = (V, E)$, there exists a preference profile R_N such that the C1 graph with respect to R_N is G. Further, the same profile yields a margin graph (the same graph) with edge weights of* 1.

Proof. Gap: TODO

□

Definition 4.2.14 (Feedback Arc Set)**.** The problem *Feedback Arc Set* (FAS) is the problem: is it possible to make a directed graph acyclic by removing k edges.

Theorem 4.2.15. *FAS is NP-Complete (Karp, 1972). FAS is NP-Complete when restricted to tournament graphs (Alon, 2006; Charbit et al. 2007; Conitzer 2006), refer to this problem as **T-FAS**.*

Theorem 4.2.16 (Bartholdi et al., 1989)**.** *Deciding whether a there exists a ranking with Kemeny score at least s is NP-Complete.*

Proof. Slightly tricky reduction from T-FAS using McGarvey's theorem and Lemma 4.2.7. Exercise. □

Remark 4.2.17. The Kemeny score problem is at least as hard as T-FAS. Finding a Kemeny ranking is at least as hard as the Kemeny score problem, so finding a Kemeny ranking is NP-Hard. Finding a Kemeny ranking is NP-Hard in the restricted case for even $n \geq 4$ and odd $n \geq 7$. If it is NP-Hard to find a Kemeny ranking for $n = 3, 5$ is open. Deciding if a given ranking is a Kemeny ranking is CoNP-Complete (clearly). There are approximation algorithms for finding a Kemeny ranking.

[7]This final condition on $x \sim_M y$ is more of a technicality which says that if there is a tie and one of the linear orders given by h has $x \succ y$, then we can flip the order of x and y in that linear order and it will still be given by h.

5

Weighted Tournament Solutions

Comment. In this chapter, we consider C2 social choice functions.

Notation 5.0.1. Given a preference profile R_N, and alternatives $x, y \in U$, let $m(x, y) := n_{xy} - n_{yx}$ be the majority margin between x and y. Clearly $m(x, y) = -m(y, x)$, and $m(x, x) = 0$.

Definition 5.0.2 (Weighted Tournament Solution)**.** An SCF is a *weighted tournament solution* if it only depends on $\big(m(x, y)\big)_{x,y \in U}$. Often, all we need is the order among $\big(m(x, y)\big)_{x,y \in U}$ (note, this is not the case for Kemeny). Basically, this is the same as be a function of the margin graph.

Theorem 5.0.3 (Debord's Theorem)**.** *The following generalises McGarvey's Theorem.*

Let $\big(m(x, y)\big)_{x,y \in U}$ be a collection of integers with $m(x, y) = -m(y, x)$ for all $x, y \in U$, and $m(x, x) = 0$ for all $x \in U$. Then, there exists a preference profile R_N with majority margins $\big(m(x, y)\big)_{x,y \in U}$ if and only if for all $x \neq y$, $m(x, y)$ has the same parity.

The parity condition makes sense, consider first that $n_{xy} + n_{yx} = n$ and so $m(x, y) = n_{xy} - n_{yx} = 2n_{xy} - n$, which has the same parity as n.

Proof. Exercise. □

§5.1 Clones

Example 5.1.1. In 1969, Thunder Bay, Ontario, Canada, held a plurality referendum to decide its name.

- Thunder Bay: 15,870 votes
- Lakehead: 15,302 votes
- The Lakehead: 8,377 votes

This phenomena is vote splitting, or manipulation by cloning.

Plurality is especially, susceptible to this, something like instant run-off would not fall to this so easily.

Definition 5.1.2 (Clones)**.** For a preference profile R_N a subset $C \subseteq U$, s.t. $|U| > |C| \geq 2$ is a set of *clones* if for all $c, c' \in C$, $x \in U \setminus C$, and $i \in N$ it holds that $c \succ_i x$ iff $c' \succ_i x$.

Informally, a set of clones are always ranked next to each other by every voter, though the ranking among the clones may be arbitrary (clones appear as a contiguous interval in each voters preference relation).

Observation 5.1.3 (Tideman, 1987). Most SCFs are susceptible to manipulation by adding or deleting clones.

Definition 5.1.4 (Independence of Clones)**.** An SCF f satisfies *independence of clones* (IoC) if for all profiles R_N, feasible set $A \in \mathcal{F}(U)$, and set of clones $C \subseteq A$, and $c \in C$ the following two properties hold.

1. For all $a \in A \setminus C$: $a \in f(R_N, A) \iff a \in f(R_N, A \setminus \{c\})$.
2. $C \cap f(R_N, A) \neq \emptyset$ if and only if $C \cap f(R_N, A \setminus \{c\}) \neq \emptyset$.

The first property says that adding or removing a clone alternative should not affect the whether a non-clone is selected or not.

The second says that adding or removing a clone from a clone group does not affect whether or not the clone group contains a winner (note we specify clone groups of size at least 2).

Example 5.1.5. Instant-runoff satisfies independence of clones, but we are not too fond of it because it fails to satisfy many other desirable properties: it is not monotonic, nor *Condorcet consistent*.

§5.2 Ranked Pairs

Definition 5.2.1 (Ranked Pairs: Tideman, 1987)**.** The SCF *Ranked Pairs* was proposed to achieve independence of clones. It may be viewed as a greedy heuristic for finding rankings with a high Kemeny score.

Ranked Pairs is an SCF that follows more algorithmcally.

1. Order the edges of the majority graph by decreasing weight, use a tiebreaker if necessary (often we assume that their are no ties and also there are no majority ties so that there is an edge between every pair in the majority graph). Restrict to just the edges with both endpoints in the feasible set A.
2. Initialize an empty graph G on the vertex set of all feasible alternatives A.
3. Process the edges as follows in the order given by (1). If $\langle a, b\rangle$ is the next majority edge to be processed:
    - If $G \cup \{\langle a, b\rangle\}$ is acyclic, add $\langle a, b\rangle$ to G.
    - Otherwise, skip $\langle a, b\rangle$ (Alternatively, add $\langle b, a\rangle$ to G).

This gives a linear order on the feasible alternatives. Finally, return the highest ranked alternative in this order, *the ranked pairs winner.*

Clearly, this is polynomial-time. Essentially we are greedily adding edges where we can starting with the ones that give the highest greedy Kemeny score (may not be optimal).

Gap: Why does this algorithm always output something?

Theorem 5.2.2 (Tideman and Zavist, 1989)**.** *When equipped with a suitable tiebreaker, Ranked Pair satisfies IoC* [1]*.*

Given a tiebreaker, Ranked Pairs is a resolute SCF and can be computed in polynomial-time.

[1] Caveat: The tiebreaker violates neutrality and/or anonymity

Definition 5.2.3 (PUT Ranked Pairs)**.** Parallel-Universes Tiebreaking. The irresolute SCF that outputs all alternatives that can win under some tiebreaker, we then can maintain anonymity and neutrality (but annoyingly, does nto satisfy independence of clones). Computing it is NP-Hard (reduction from SAT).

Definition 5.2.4 (Stability for Winners: Holliday and Pacuit, 2023)**.** An SCF satisfies *stability for winners* if for all profiles R_N, and all feasible sets A and $b \in U \setminus A$: if $a \in f(R_N, A)$ and $a \succ_M b$, then $a \in f(R_N, A \cup \{b\})$.

Remark 5.2.5. Stability for winners implies that there can not be a *spoiler effect* as in the Gore, Bush, Nader case.

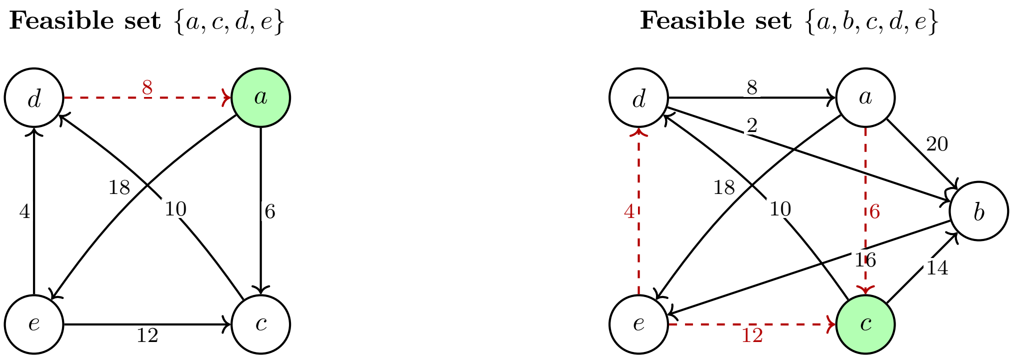


Figure 5.1: Ranked Pairs fails stability for winners. Solid arrows are the majority edges that are included in the ranked pairs algorithm; dashed red arrows are rejected edges (would create a cycle). Winner shown in green. Removing non-winner b changes the winner from c to a.

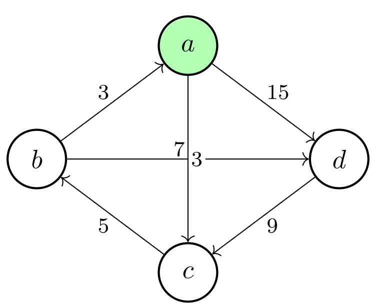
Example 5.2.6 (Ranked Pairs Fails Stability for Winners)**.**

Definition 5.2.7 (The (strong) no show paradox)**.** Consider the following situation. A profile R_N has that x is a winner under f. A voter $i \notin N$ ranks x first. The new profile $R_{N \cup \{i\}}$ no longer gives that x is the winner.

It is a different type of manipulability, where it would be beneficial for a voter to not participate in an election.

This is known as the *(strong) no show paradox*.

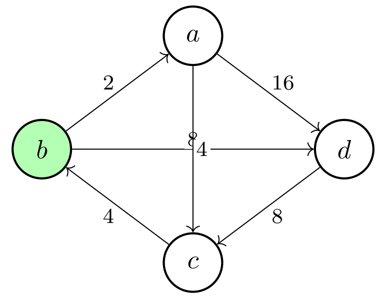
Example 5.2.8 (Ranked pairs is susceptible to the no show paradox)**.** Consider the following margin graph.



The ranked pairs algorithm will select all but the two 3 edges, and give that a is the winner.

Consider the profile now with a new voter i with preference relation $a \succ_i b \succ_i c \succ_i d$.

The new profile looks like,



And we have winner, b.

Definition 5.2.9 (Positive Involvement)**.** An SCF f satisfies positive involvement if for all feasible sets A and alternatives $x \in A$, and all profiles R_N and $R_{N'}$ where $N' = N \cup \{i\}$, where $\text{Max}(\succsim_i, A) = x$: if $x \in f(R_N, A)$, then $x \in f(R_{N'}, A)$.

§5.3 Split Cycle

Definition 5.3.1. Consider a preference profile and let $C = (x_1, \ldots, x_k)$ be a simple cycle in the margin graph [2], when restricted to a feasible set A. The splitting number $\text{sp}(C)$ is the smallest margin between consecutive alternatives in C.

We say that a *defeats* b if $a \succ_M b$ and for every (simple) cycle C containing a and b, $\text{sp}(C) < m(a, b)$.

Informally, we think that a defeats b if it wins it strictly wins in the majority relation, and further the edge from $a \to b$ is never the weakest link on a cycle.

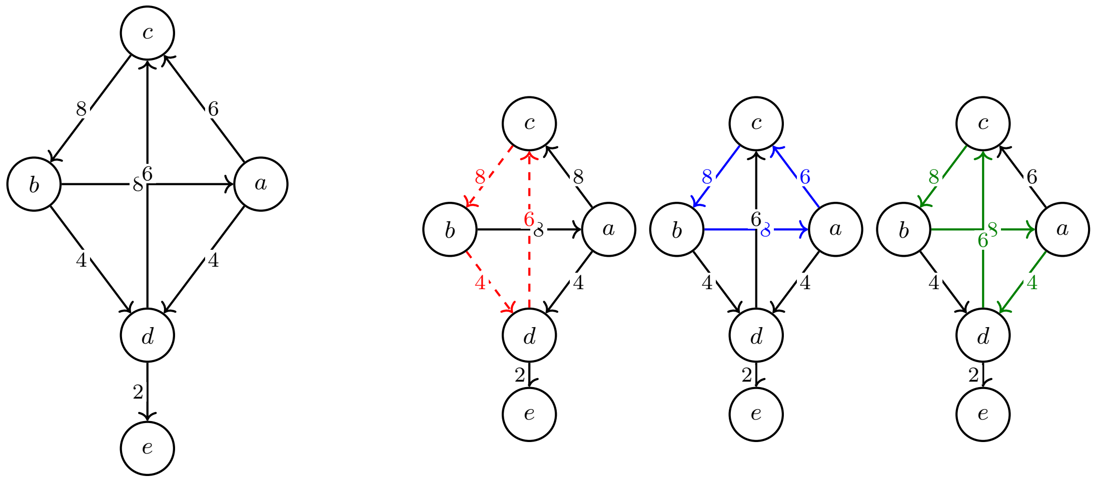
Definition 5.3.2 (Split Cycle)**.** The social choice function *Split Cycle* is the function that returns all undefeated candidates.

Denote it by f_{SC}.

We can think of the computation of split cycle as simply starting with the margin graph, identifying every simple cycle in the margin graph and deleting the lightest edge along each cycle, finally it outputs those alternatives with no incoming edges exactly.

Example 5.3.3. Consider the following margin graph, in colours each simple cycle is highlighted. The Split Cycle algorithm would remove those edges corresponding to the lightest weight in the cycle, and then output the candidates that have no incoming edges — these are precisely the undefeated candidates. The remaining edges give exactly the defeat relation.

[2]A path as opposed to a walk.



Red: cycle (c, b, d), sp $= 4$ Blue: cycle (c, b, a), sp $= 6$ Green: cycle (c, b, a, d), sp $= 4$

It is clear, that the defeat relation is **not** transitive, since after deleting these three lightest edges, c defeats b and b defeats a, but c does not defeat a.

Lemma 5.3.4 (Defeat Relation is acyclic)**.** *We can think of* defeat *as a relation on alternatives. To show that Split Cycle gives a non-empty choice set, it suffices (but is not necessary) to show that the defeat relation is acyclic.*

Proof. Suppose for contradiction that there is a cycle under the defeat relation, then there is a simple cycle under the the defeat relation, say $a_1, \ldots, a_k$. Then, by definition of the defeat relation, $C := a_1, \ldots, a_k$ is a simple cycle in the margin graph. Then, consider that for each edge $a_i \to a_{i+1}$ (incl. the edge $a_k \to a_1$), in the defeat cycle, by definition of defeat, we have that $m(a_i, a_{i+1}) > \mathrm{sp}(C)$. Consider that by definition of splitting number, there is some edge $a_j \to a_{j+1}$ in C such that $m(a_j, a_{j+1}) = \mathrm{sp}(C)$, contradicting the fact that $m(a_j, a_{j+1}) > \mathrm{sp}(C)$. $\square$

Observation 5.3.5. To check whether a defeats b, we only need to check the simple cycles containing the edge $a \to b$. That is we need not check cycles containing a and b when they are not adjacent, and certainly not all simple cycles in the margin graph. We omit proof.

Definition 5.3.6. For alternatives $a, b \in U$ with $a \succ_M b$, define the *cycle number* to be

$$\mathrm{cyc}(a, b) := \max \Big(\{0\} \cup \{\mathrm{sp}(C) : \text{C a simple cycle in margin graph containing edge}\langle a, b\rangle\} \Big)$$

Informally, the cycle number of a and b is the the maximum splitting number among simple cycles containing the edge $a \to b$, or 0 if there are no such cycles.

Lemma 5.3.7. *a defeats b if and only if $m(a, b) > \mathrm{cyc}(a, b)$.*

Proof. Exercise. $\square$

Remark 5.3.8. The naive approach we give to compute f_{SC} requires checking all simple cycles in the margin graph, which is exponential in the number of alternatives. We hope to use the above ideas, to get a more computationally appealing method to compute f_{SC}.

Definition 5.3.9. The *strength* of a path in the margin graph is the smallest margin between consecutive alternatives on the path.

Let $\mathrm{strength}(x, y)$ denote the strength of the strongest path from x to y.

Lemma 5.3.10. *a defeats b if and only if $a \succ_M b$ and* $\text{strength}(b,a) < m(a,b)$*.*

Remark 5.3.11. We characterise the notion of defeat via strength because $\text{strength}(\cdot,\cdot)$ is a notion that *can* be computed efficiently using a variant of the Floyd-Warshall algorithm ($O(|A|^3)$). Thus, we can compute split cycle winners in polynomial time. Explicitly we do this by,

1. Compute the margin graph from the profile: $O(|A|^2 n)$
2. Compute for each pair of alternatives $\text{strength}(a,b)$: $O(|A|^3)$ each so $O(|A|^5)$ total.
3. For each pair, check if a defeats b by checking if $a \succ_M b$ and $\text{strength}(a,b) < m(a,b)$: $O(|A|^2)$.
4. Return all those alternatives with no incoming edges in the defeat graph.

Proposition 5.3.12 (Split Cycle satisfies stability for winners)**.**

Proof. Suppose for contradiction that split cycle fails stability for winners, then there is a profile R_N, feasible set A, and $b \in U \setminus A$ such that there is some $a \in f_{\text{SC}}(R_N, A)$ and $a \succ_M b$, but $a \notin f_{\text{SC}}(R_N, A \cup \{b\})$.

Consider that adding b to the set of feasible alternatives can only add paths in the margin graph between vertices, so $\text{strength}_{A\cup\{b\}}(x,y) \geq \text{strength}_A(x,y)$ for all $x, y \in A$. For contradiction, suppose a is defeated in $A \cup \{b\}$, then there is some alternative c such that $\text{strength}_{A\cup\{b\}}(a,c) < m(c,a)$. But $\text{strength}_{A\cup\{b\}}(a,b) \geq \text{strength}_A(a,b)$, thus $\text{strength}_A(a,c) < m(c,a)$. If $c \in A$, then a is defeated in A, contradicting that $a \in f_{\text{SC}}(R_N, A)$. Thus, $c = b$, but this gives a cyclic strict majority between a and b, since $a \succ_M b$, giving contradiction. □

Proposition 5.3.13 (Split Cycle satisfies positive involvement)**.**

Proof. Exercise. □

Proposition 5.3.14 (Split Cycle satisfies independence of clones)**.**

Proof. Omitted. □

Proposition 5.3.15 (Split Cycle subsumes ranked pairs)**.** *If f_{RP} is the ranked pairs SCF, then for all profiles R_N and feasible sets A, $f_{RP}(R_N, A) \subseteq f_{SC}(R_N, A)$.*

Sketch. We omit the details of tie-breaking.

Suppose that $a \notin f_{\text{SC}}(R_N, A)$, then there is some b that defeats a. Then, Lemma 5.3.10's gives that $b \succ_M a$ or $\text{strength}(b,a) < m(a,b)$. Now consider the cycle formed by the traversal from $b \to_M a$ and then along the strongest path from a to b. At some point the ranked pairs algorithm will encounter the edge $b \to a$ and will add it because, all cycles containing $b \to a$ have splitting number less than $m(a,b)$, so the ranked pairs algorithm will never reject the edge $b \to a$, and thus a can not be the winner under ranked pairs. □

Comment. To understand the trade-offs between Ranked Pairs and Split Cycle, we briefly look at the following impossibility results.

First, note that both Ranked Pairs and Split Cycle satisfy *ordinal margin invariance*; that is they depend only on the ordering of the margins $(m(x,y))_{x,y\in U}$, not on the actual values of the margins. Note, Kemeny does care about the numbers not just their ordering.

Theorem 5.3.16 (Holliday, 2023)**.** *The following are incompatible: Condorcet extension, Condorcet loser criterion, ordinal margin invariance, positive involvement, and resolvability.*

Where by Condorcet loser criterion *we mean that we never select a Condorcet loser. And, by* resolvability *we mean that for all profiles R_N and feasible sets A, when $|f(R_N, A)| > 1$, there is some profile $R_{N\cup\{i\}}$ such that $|f(R_{N\cup\{i\}}, A)| = 1$ (we can always add one voter to break a tie).*

Remark 5.3.17. Split Cycle satisfies all but resolvability, while Ranked Pairs satisfies all but positive involvement.

Definition 5.3.18 (Quasi-Resoluteness)**.** An SCF f is *quasi-resolute* if for all profiles R_N and feasible sets A, where the edges of the margin graph are all distinct, then $|f(R_N, A)| = 1$.

Such profiles are sometimes called *uniquely weighted*. It is clear that they do not permit majority ties, i.e. for all $x \neq y$, $m(x, y) \neq 0$.

Theorem 5.3.19 (Holliday et al., 2024)**.** *There is no anonymous and neutral SCF that satisfies stability for winners and quasi-resoluteness.*

6

Multiwinner Voting Rules

Idea 6.0.1. What does multiwinner votine mean? So far, we have modelled alternatives as mutually exclusive outcomes, and we wanted to the chose the *best* alternative, or rank all alternatives. Of course, many rules we have discussed are irresolute, but this is not exactly desired behavior — our framework ushered us away from multiple winners (implicitly we imagine some out of framework tie breaker).

What if instead, we imagine we *must* elect a a specifically sized (k) set of of alternatives.

In principle, we could model multiwinner elections with the tools we have thus far, we could have voters just rank their preferences where candidates are different k-subsets of alternatives (different committees) — but this gives exponential number of alternatives, and thus is not computationally appealing, nor natural.

Another idea is to use a social welfare function, and pick the top k alternatives in the social welfare ordering, we revisit this later.

Remark 6.0.2. We consider two different frameworks for multiwinner voting, whereby voters *indirectly* express preferences over committees.

- The *Apportionment setting*: candidates are grouped into *parties* and voters can support a single party only. The underlying assumption is that: voters only care about how many candidates from their party are selected.

  This models, *party-list election*, which are often used in parliamentary elections. Intuitively, it seems clear that it kind of limits the expressive power of voters.

- The *Approval-Based setting*, voters can approve whichever candidates, but they do not rank them (still less expressive than our previous settings). The assumption is that voters only care about how many candidates that they approve of are selected.

In both settings, we are interested in a notion of *proportional representation.*

§6.1 Apportionment Setting

Definition 6.1.1. In the apportionment setting, we have a set of parties $P = \{P_1, \ldots, P_p\}$, and each voter votes for exactly one party.

For each party P_i let v_i denote the number of votes it recieves. The goal is to allocate a given number of seats among the parties, *proportional* to the votes.

Gap: TODO: Crib sheet for all the rules and the properties that they satisfy.